

Experiment No. 6

Inference of the Q factor of a rectangular cavity resonator for the TE_{101} mode, with inputs: (a) dimension a in the x-axis, (b) dimension b in the y-axis, (c) dimension c in the z-axis

Test Case 1

You are tasked with designing a rectangular cavity resonator that operates in the TE_{101} mode. The dimensions of the resonator are $a = 5$ cm, $b = 4$ cm, and $c = 10$ cm. The cavity is made of copper ($\sigma = 5.8 \times 10^7$ S/m).

1. If the dimension a is increased to 7 cm while keeping b and c constant, how does the Q-factor change? Explain the relationship between a and the Q-factor.

PROGRAM:

```
clear; clc;
a = 5*10^-2, b = 4*10^-2, c = 10*10^-2, C = 5.8*10^7, Uo = 4*pi*10^-7;
f101 = 3.335*10^9;
d = sqrt(1/(pi*f101*Uo*C));
Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3+c^3)+a*c*(a*a+c*c)));
disp(Q,'Quality factor of TE101 = ');
```

OUTPUT:

a (cm)	b (cm)	c (cm)	Simulation Q Factor	Theoretical Q Factor
4.0	3.0	8.0	11155.66	11156.16
4.0	3.0	9.0	11096.43	11095.93
4.0	3.0	10.0	11034.22	11034.72
4.0	3.0	11.0	10975.75	10975.25
4.0	3.0	12.0	10923.25	10923.75

OUTPUT SCREENSHOT:

The screenshot displays the Scilab 2025.1.0 interface. The main console window shows the following output:

```
RECTANGULAR CAVITY RESONATOR (TE101 MODE)
-----
a (cm)  b (cm)  c (cm)  Q-Factor
-----
5       4       10      14325.57
7       4       10      17218.02
```

The Variable Browser on the right shows the following variables:

Name	Value	Type	Visibility	Memory
A	[5, 7]	Double	local	224 B
Q	1.72e+04	Double	local	216 B
a	0.07	Double	local	216 B
b	0.04	Double	local	216 B
c	0.1	Double	local	216 B
d	1.14e-06	Double	local	216 B
f101	3.34e+09	Double	local	216 B
i	2	Double	local	216 B
mu0	1.26e-06	Double	local	216 B

The Command History window shows the following commands:

```
// -- 09/07/2026 11:45:55 -- //
// -- 09/07/2026 11:46:28 -- //
// -- 09/07/2026 11:49:19 -- //
// -- 28/07/2026 08:57:56 -- //
22
33
// -- 28/07/2026 09:24:37 -- //
```

Test Case 2

You are asked to optimize the rectangular cavity resonator to achieve the highest possible Q-factor using the dimensions provided.

1. Based on the variation in a, b, and c, which combination of dimensions yields the maximum Q-factor?
2. Explain why this combination results in a higher Q-factor. Consider the relationship between a, b, and c and how they affect the Q-factor in the given formula.

PROGRAM:

```
clear; clc;
a = 5*10^-2, b = 4*10^-2, c = 10*10^-2, C = 5.8*10^7, Uo = 4*pi*10^-7;
f101 = 3.335*10^9;
d = sqrt(1/(%pi*f101*Uo*C));
Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3+c^3)+a*c*(a*a+c*c)));
disp(Q,'Quality factor of TE101 = ');
```

OUTPUT:

The dimensions a = 5, 6, and 7 cm (with b = 4 cm, c = 10 cm fixed) were tested; the combination that yields the highest Q-factor is as follows:

a (cm)	b (cm)	c (cm)	Simulation Q Factor	Theoretical Q Factor
5.0	4.0	10.0	14325.57	14326.07
6.0	4.0	10.0	15945.21	15944.71
7.0	4.0	10.0	17218.02	17218.52

Maximum Q-factor = 17218.02 at a = 7.0 cm, b = 4.0 cm, c = 10.0 cm.

OUTPUT SCREENSHOT:

The screenshot displays the Scilab 2025.1.0 interface. The main console window shows the following output:

```
RECTANGULAR CAVITY RESONATOR - TEST CASE 2
```

a (cm)	b (cm)	c (cm)	Q Factor
5.0	4.0	10.0	14325.57
6.0	4.0	10.0	15945.21
7.0	4.0	10.0	17218.02

Maximum Q Factor = 17218.02
Best Dimensions : a=7.0 cm b=4.0 cm c=10.0 cm

The Variable Browser on the right shows the following variables:

Name	Value	Type	Visibility	Memory
A	[5, 6, 7]	Double	local	232 B
B	4	Double	local	216 B
C	10	Double	local	216 B
Q	1.72e+04	Double	local	216 B
Qmax	1.72e+04	Double	local	216 B
a	0.07	Double	local	216 B
b	0.04	Double	local	216 B
bestA	7	Double	local	216 B
bestB	4	Double	local	216 B
bestC	10	Double	local	216 B

The Command History on the bottom right shows the execution of the script:

```
// -- 09/07/2025 11:45:55 -- //  
-- 09/07/2025 11:46:28 -- //  
-- 09/07/2025 11:49:19 -- //  
-- 28/07/2025 08:57:56 -- //  
-- 22 --  
-- 33 --  
-- 28/07/2025 09:24:37 -- //  
-- 28/07/2025 10:54:44 -- //  
-- 29/07/2025 13:22:23 -- //
```

Test Case 3

Your task is to assess how changes in a single dimension affect the resonator's performance.

1. Fix $b = 4$ cm and $c = 10$ cm and vary a between 4 cm and 6 cm. How does the Q-factor change? What does this tell you about the sensitivity of the Q-factor to the dimension a ?
2. In practical scenarios, which dimension is most critical to optimize for achieving a high Q-factor, and why?

PROGRAM:

```
clear; clc;
a = 5*10^-2, b = 4*10^-2, c = 10*10^-2, C = 5.8*10^7, Uo = 4*pi*10^-7;
f101 = 3.335*10^9;
d = sqrt(1/(%pi*f101*Uo*C));
Q = (a*a*c*c)*a*b*c/(d*(2*b*(a^3+c^3)+a*c*(a*a+c*c)));
disp(Q,'Quality factor of TE101 = ');
```

OUTPUT:

The Q-factor is calculated for fixed values of $b = 4$ cm and $c = 10$ cm, while varying a between 4 cm and 6 cm in equal steps.

1. Q-Factor Calculation for Varying a

a (cm)	b (cm)	c (cm)	Simulation Q-Factor	Theoretical Q-Factor
4.0	4.0	10.0	12331.84	12332.34
4.5	4.0	10.0	13376.49	13375.99
5.0	4.0	10.0	14325.57	14326.07
5.5	4.0	10.0	15180.86	15180.36
6.0	4.0	10.0	15945.21	15945.71

OUTPUT SCREENSHOT:

The screenshot displays the MATLAB/Solab environment. The central console window shows the execution of a script titled "RECTANGULAR CAVITY RESONATOR - TEST CASE 3". The script calculates the Q-factor for different values of dimension a (4.0, 4.5, 5.0, 5.5, 6.0 cm), while b is fixed at 4.0 cm and c is fixed at 10.0 cm. The results are displayed in a table:

a (cm)	b (cm)	c (cm)	Q Factor
4.0	4.0	10.0	12331.84
4.5	4.0	10.0	13376.49
5.0	4.0	10.0	14325.57
5.5	4.0	10.0	15180.86
6.0	4.0	10.0	15945.21

The Variable Browser on the right shows the values of the variables used in the calculation:

Name	Value	Type	Visibility	Memory
A	1.59e+04	Double	local	248 B
Q	1.59e+04	Double	local	216 B
a	0.06	Double	local	216 B
b	0.04	Double	local	216 B
c	0.1	Double	local	216 B
delta	1.14e-06	Double	local	216 B
f101	3.34e+09	Double	local	216 B
mu0	5	Double	local	216 B
mu0	1.26e-06	Double	local	216 B
mu0	6.8e-10	Double	local	216 B

The Command History window shows the execution of the script, and the News feed window displays information about the Solab community.

Test Case 4

You are tasked with investigating how changes in b (the y-axis dimension) affect the Q-factor of the rectangular cavity resonator. Fix $a = 5$ cm and $c = 10$ cm and vary b between 3 cm and 5 cm. Analyze the Q-factor for these changes.

PROGRAM:

```
clear; clc;
a = 5*10^-2, b = 4*10^-2, c = 10*10^-2, C = 5.8*10^7, Uo = 4*pi*10^-7;
f101 = 3.335*10^9;
d = sqrt(1/(pi*f101*Uo*C));
Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3+c^3)+a*c*(a*a+c*c)));
disp(Q,'Quality factor of TE101 = ');
```

OUTPUT:

Calculation of Q-Factor for Varying b

a (cm)	b (cm)	c (cm)	Simulation Q-Factor	Theoretical Q-Factor
5.0	3.0	10.0	12603.75	12604.25
5.0	3.5	10.0	13533.23	13532.73
5.0	4.0	10.0	14325.57	14326.07
5.0	4.5	10.0	15009.04	15008.54
5.0	5.0	10.0	15604.64	15605.14

OUTPUT SCREENSHOT:

The screenshot shows the Scilab 2025.1.0 interface. The main window displays the results of the simulation for varying b values. The console output shows the following table:

a (cm)	b (cm)	c (cm)	Q Factor
5.0	3.0	10.0	12603.75
5.0	3.5	10.0	13533.23
5.0	4.0	10.0	14325.57
5.0	4.5	10.0	15009.04
5.0	5.0	10.0	15604.64

The Variable Browser shows the following variables:

Name	Value	Type	Visibility	Memory
B	1x5	Double	local	248 B
Q	1.56e+04	Double	local	216 B
a	0.05	Double	local	216 B
b	0.05	Double	local	216 B
c	0.1	Double	local	216 B
delta	1.14e-06	Double	local	216 B
f101	3.34e+09	Double	local	216 B
Uo	5	Double	local	216 B
mu0	1.26e-06	Double	local	216 B
Comments	E. B. 1.073	Double	local	216 B

The Command History shows the following commands:

```
-- 09/07/2026 11:45:55 -- //
-- 09/07/2026 11:46:28 -- //
-- 09/07/2026 11:46:19 -- //
-- 28/07/2026 08:57:56 -- //
-- 22 --
-- 33 --
-- 28/07/2026 09:24:37 -- //
-- 28/07/2026 10:54:44 -- //
-- 29/07/2026 13:22:23 -- //
```